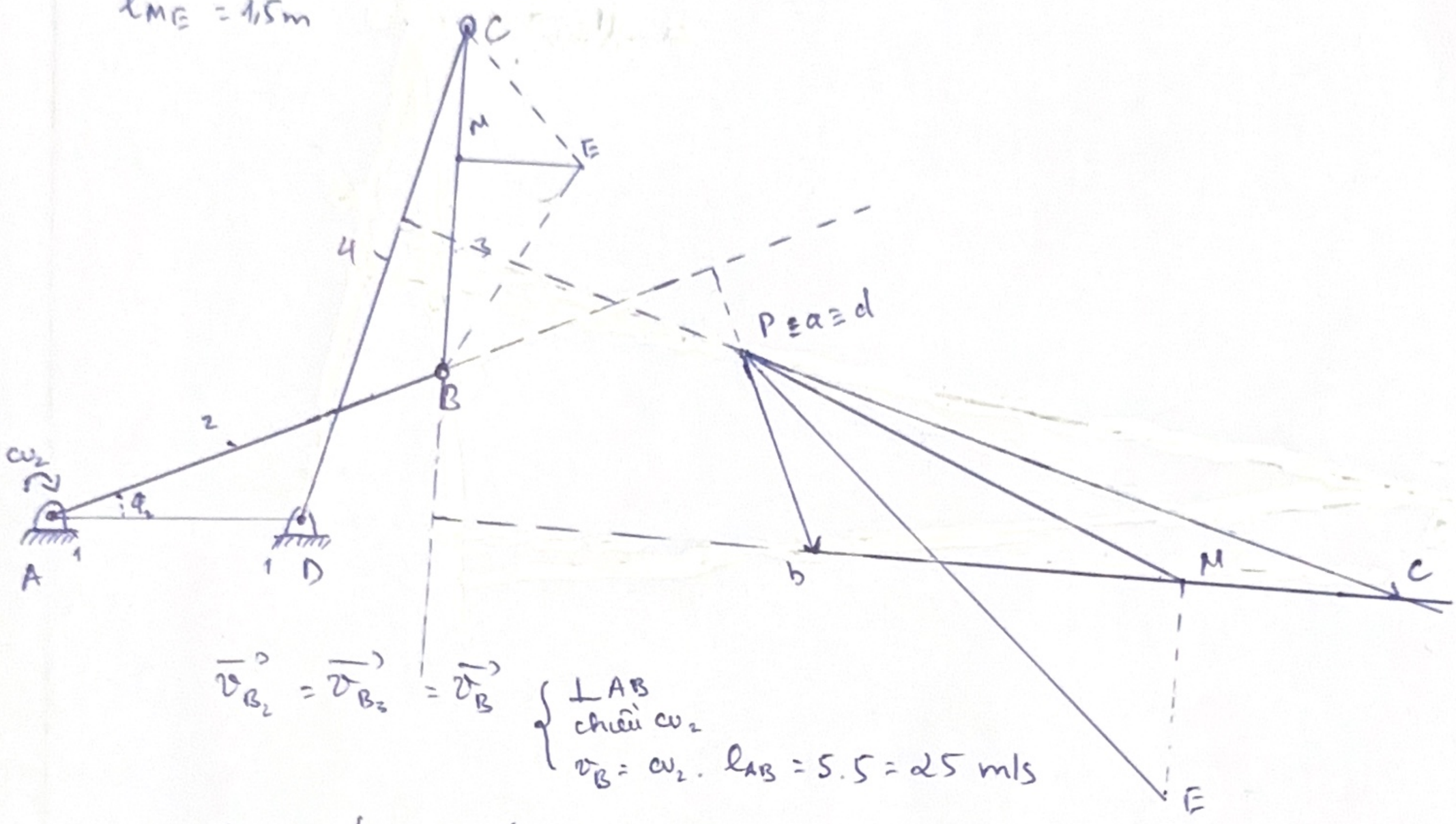
Phương án 2

$l_{AB} = 5m$ $l_{CD} = 6m$ $\varepsilon_2 = 20\ rad/s^2$ ↺

$l_{BC} = 4m$ $l_{AD} = 3m$ $\omega_2 = 5\ rad/s$ ↻

$l_{BM} = 2,5m$ $\varphi_2 = 20°$

$l_{ME} = 1,5m$



$\vec{v}_{B_2} = \vec{v}_{B_3} = \vec{v}_B$ $\begin{cases} \perp AB \\ \text{chiều } \omega_2 \\ v_B = \omega_2 . l_{AB} = 5.5 = 25\ m/s \end{cases}$

Chọn gốc vận tốc p

Dùng $\vec{pb}$ để biểu diễn $\vec{v_B}$

Chọn $|pb| = 25\ mm \Rightarrow \mu_v = \frac{v_B}{pb} = \frac{1}{} \frac{m/s}{mm}$

+) Phút tốc C

$\vec{v_C} = \vec{v_B} + \vec{v_{CB}}$

$\begin{cases} \perp CD \\ ? \\ l_{CD}.\omega_4 \end{cases}$ $\begin{cases} \vdots \\ \vdots \\ \vdots \end{cases}$ $\begin{cases} \perp BC \\ \\ \omega_3 . l_{BC} \end{cases}$ $\Rightarrow \vec{pc} = \vec{pb} + \vec{bc}$

$v_c = |pc| . \mu_v = 85 . 1 = 85\ m/s \Rightarrow \omega_4 = \frac{85}{6} \approx 14,2\ rad/s$ cùng chiều ω_2

$v_{BC} = |bc| . \mu_v = 72 . 1 = 72\ m/s \Rightarrow \omega_3 = \frac{v_{BC}}{l_{BC}} \approx 18\ rad/s.$ cùng chiều ω_2

$\frac{BM}{BC} = \frac{bm}{bc} \Rightarrow \frac{bm}{bc} = \frac{2,5}{4} = \frac{5}{8} \Rightarrow bm = 45mm$

$v_M = |pm| . \mu_v = 59,4 . 1 = 59,4\ m/s$

$\Delta BCE \backsim \Delta bce$ ($me \perp bc$) $\Rightarrow \frac{ME}{BC} = \frac{me}{bc} \Rightarrow me = bc . \frac{ME}{BC} = 27mm$

$\Rightarrow pe \approx 74,4\ mm$

$\Rightarrow v_E = |pe| \mu_v = 74,4\ m/s$